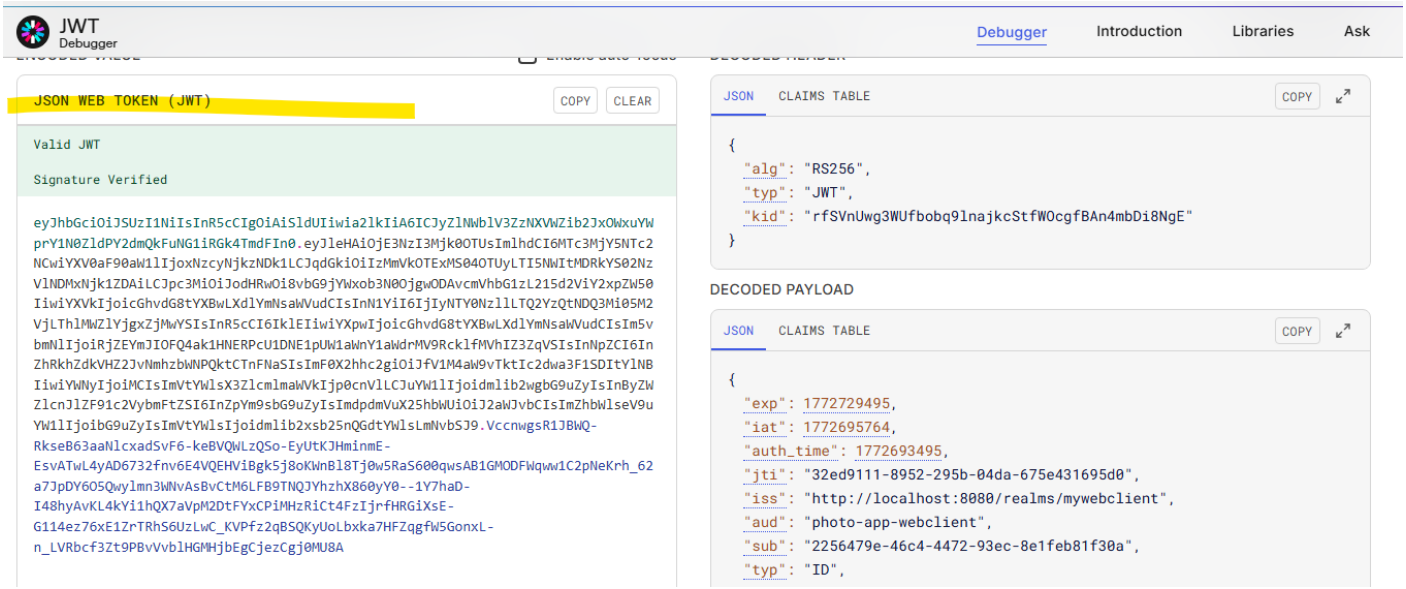
How to get JWT Access Tokeng
(TokenValue From OidcUser principal)

ពេលដែលយើង login success ហើយ វានឹងបាន principal from Auth Server មកវិញ ចឹងសម្រាប់ Spring វាធ្វើការអោយយើងហើយ ដោយវាដឹងថាទៅយកពី Keyload or google server or facebook ចឹងយើងអាច យកTokenValue នឹងមកបាន




```java
@Autowired
OAuth2AuthorizedClientService oauth2ClientService;

@GetMapping("/albums")
public String getAlbums(Model model,
        @AuthenticationPrincipal OidcUser principal) {

    Authentication authentication = SecurityContextHolder.getContext().getAuthentication();
    OAuth2AuthenticationToken oauthToken = (OAuth2AuthenticationToken) authentication;

    OAuth2AuthorizedClient oauth2Client = oauth2ClientService.loadAuthorizedClient(oauthToken.getAuthc
            oauthToken.getName());

    String jwtAccessToken = oauth2Client.getAccessToken().getTokenValue();
    System.out.println("jwtAccessToken = " + jwtAccessToken);


    System.out.println("Principal = " + principal);

    OidcIdToken idToken = principal.getIdToken();
    String idTokenValue = idToken.getTokenValue();
    System.out.println("idTokenValue = " + idTokenValue);
```

1- We use Authentication object to get the JWT token យើងធ្វើការ injected ទៅ class ដែលយើងធ្វើការ Authenticated នឹងតែម្តង់

Explain:

យើងអាចធ្វើការ Inject យក Object បានតាមរយៈដាក់វាចូលក្នុង parameter វាបាន public String getAlbums(Model model, @AuthenticationPrincipal OidcUser principal, Authentication authentication)

ឬយើងអាចធ្វើការទាញ យក Authentication object ដោយធ្វើការបង្កើតវានៅខាងក្រៅបាន (SecurityContextHolder.getContext().getAuthentication();)

យើងអាចអោយវា Retrun object Authentication មកយើងអាចលុយ injection វិញបានហើយ

```
// from spring security we can get the Authentication from SecurityContextHolder
Authentication authentication = SecurityContextHolder.getContext().getAuthentication();
```

authentication វាមានផ្ទុកនូវ JWT ក្នុងនឹង

2- យើងត្រូវបំលែងវាពី OAuth ទៅកាន់ Jave Authentication

2) What is `OAuth2AuthenticationToken`?

In Spring Security, when the user signs in using **OAuth2 Login** (Google, GitHub, Okta, etc.), Spring puts an `Authentication` object into the `SecurityContext`.

If that login is OAuth2-based, the `Authentication` is often an `OAuth2AuthenticationToken`, which holds:

- the **principal** (user attributes)
- **authorities**
- the **registrationId** (which provider: google/github/...)
- and is associated with an `OAuth2AuthorizedClient` (where tokens like access token live)

1) OAuth (and OAuth 2.0) vs "AuthenticationToken"

OAuth / OAuth 2.0

- **OAuth 2.0 is an authorization framework**: it's the protocol/flow that lets a user grant an app access to resources (scopes) without sharing their password.
- OAuth 2.0 defines concepts like **authorization code flow**, **access token**, **refresh token**, **scopes**, **client id/secret**, etc.

AuthenticationToken (in Spring Security)

- `AuthenticationToken` is **a Java/Spring Security type** (a class/interface family) used **inside your application** to represent an authenticated (or not-yet-authenticated) security identity.
- Example Spring tokens:
  - `UsernamePasswordAuthenticationToken` (form login)
  - `BearerTokenAuthenticationToken` (incoming JWT bearer token)
  - `OAuth2AuthenticationToken` (user logged in via OAuth2/OIDC)
- So: **OAuth2 is a protocol**, while **AuthenticationToken is your app's internal representation** of "who is this user and what authorities do they have?"

- ចាំ OAuth 2.0 វាជា framework ដែលអាចអោយ User ទទួលបាននូវ សិទ្ធក្នុងការប្រើប្រាស់ Resource
- AuthenticationToken (spring security) វាជាប្រភេទសរបស់ Java ដែលធ្វើការទៅលើកសុវត្ថិភាពដើម្បីផ្ទាល់ផ្ទាត់អ្នកប្រើប្រាស់ពេល Login ចូល app

3- ប្រើ Sevice របស់ Spring framwork ដើម្បីធ្វើការយក JWT token

```
@Autowired
    OAuth2AuthorizedClientService oauth2ClientService;
    // this client service
```

```
OAuth2AuthorizedClient oauth2Client =
oauth2ClientService.loadAuthorizedClient(oauthToken.getAuthorizedClientRegistrationId(), oauthToken.getName());
```

យកអាចយក ទិន្នន័យពី client ដែលធ្វើការ login នឹងបានហើយ ដែល Return type វាជា OAuth2AuthorizedClient វាត្រូវការ ៖

oauthToken.getAuthorizedClientRegistrationId() ដែលបានមក ពី OAuth2AuthenticationToken oauthToken = (OAuth2AuthenticationToken) authentication; និង ត្រូវការ (), oauthToken.getName() ឈ្មោះអ្នក Login ចូលនឹង

⇨ យើងធ្វើការ យក JWT Token ដោយ ប្រើ oauth2Client

What we want:

String jwtAccessToken = oauth2Client.getAccessToken().getTokenValue();

```
@Controller
public class AlbumsController {

    @Autowired
    OAuth2AuthorizedClientService oauth2ClientService;
    // this client service

    @GetMapping("/albums")
    public String getAlbums(Model model, @AuthenticationPrincipal OidcUser principal) {

        // from spring security we can get the Authentication from SecurityContextHolder
        Authentication authentication = SecurityContextHolder.getContext().getAuthentication();

        OAuth2AuthenticationToken oauthToken = (OAuth2AuthenticationToken) authentication;
        //it means: “Treat auth as an OAuth2AuthenticationToken.”

//      1-step get the oath2 object
        OAuth2AuthorizedClient oauth2Client = oauth2ClientService.loadAuthorizedClient(oauthToken.getAuthorizedClientRegistrationId(),
                oauthToken.getName());

        String jwtAccessToken = oauth2Client.getAccessToken().getTokenValue();

        System.out.println("*** jwtAccessToken = " + jwtAccessToken + "***");
```